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L1 The Atmosphere Around You

August 179, 2025

TROPOSPHERE

It's where we are! It's where weather happens. 80% of matter in the atmosphere

STRATOSPHERE

Ozone layer protects us from UV

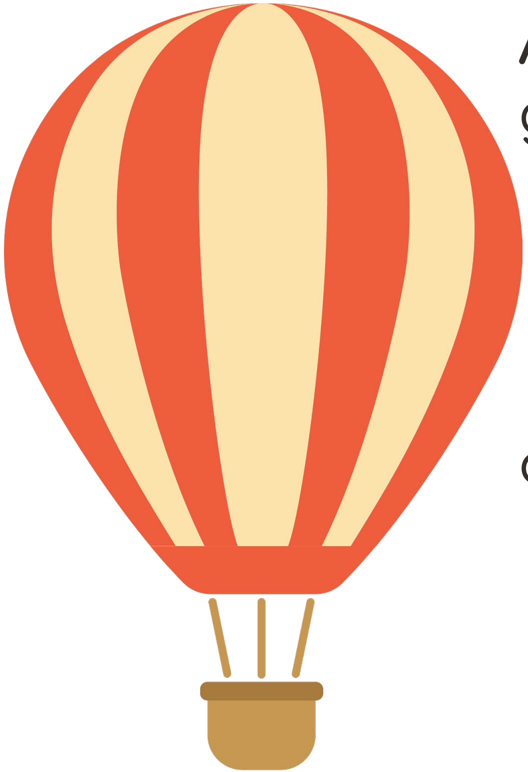
MESOSPHERE

Meteors burn up :(

THERMOSPHERE

Space starts there??? Auroras

Very hot particles because of radiation



Air Pressure:

generally decreases as altitude increases

- pressure is caused by having air pulled down by gravity, so the higher the altitude, the less air there is above to be pressing down

different air masses can have different pressures depending on a few factors

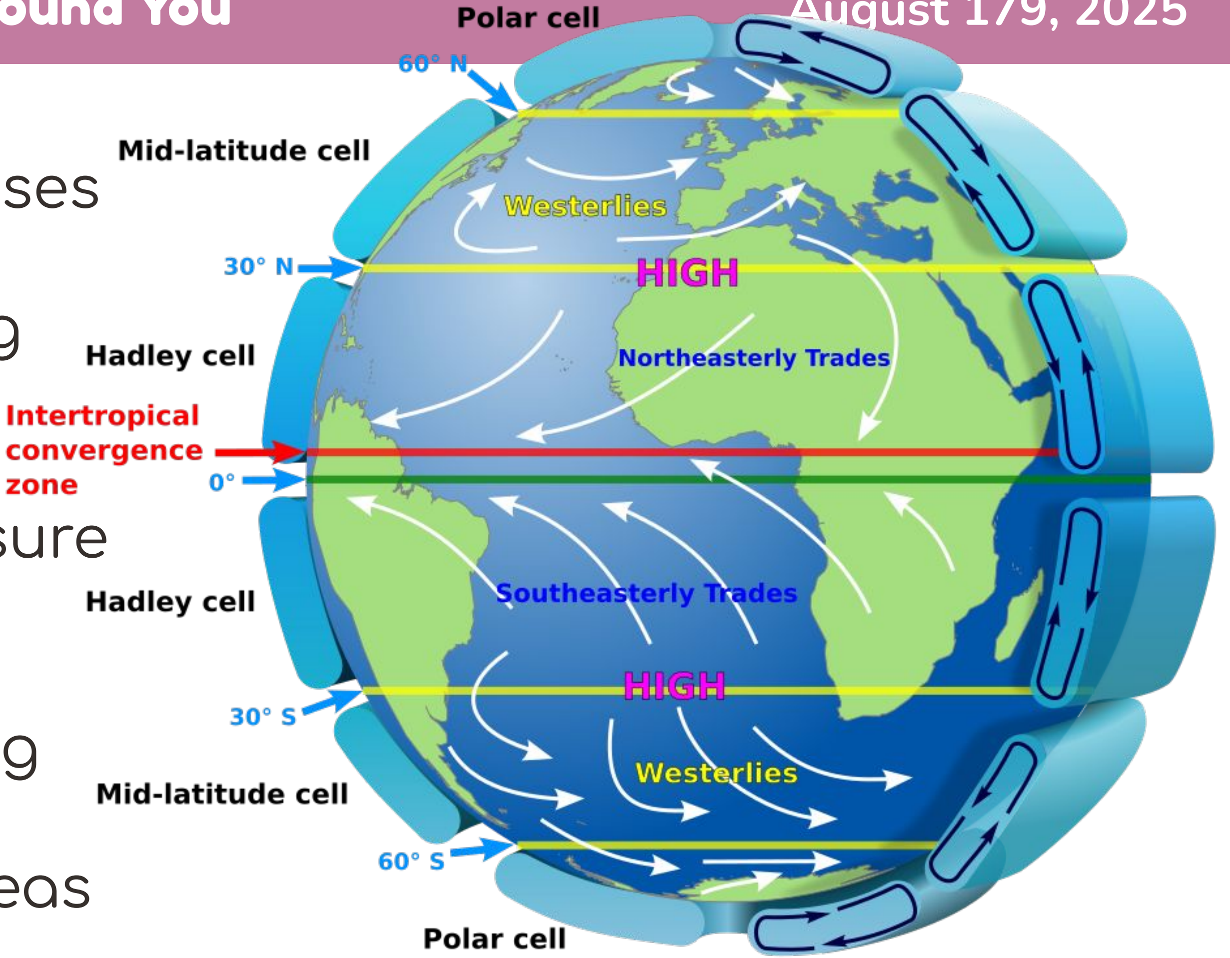
- hot air has lower pressure than cold air
- cold air is dryer and more dense

Wind:

caused by air masses of different pressures moving

high pressure air masses move to replace low pressure

the surface of the earth is heated unevenly, creating low pressure and high pressure areas

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WHERE DOES
WIND COME FROM?

NWS Heat Index

Temperature (°F)

Relative Humidity (%)

	80	82	84	86	88	90	92	94	96	98	100	102	104	106	108	110
40	80	81	83	85	88	91	94	97	101	105	109	114	119	124	130	136
45	80	82	84	87	89	93	96	100	104	109	114	119	124	130	137	
50	81	83	85	88	91	95	99	103	108	113	118	124	131	137		
55	81	84	86	89	93	97	101	106	112	117	124	130	137			
60	82	84	88	91	95	100	105	110	116	123	129	137				
65	82	85	89	93	98	103	108	114	121	128	136					
70	83	86	90	95	100	105	112	119	126	134						
75	84	88	92	97	103	109	116	124	132							
80	84	89	94	100	106	113	121	129								
85	85	90	96	102	110	117	126	135								
90	86	91	98	105	113	122	131									
95	86	93	100	108	117	127										
100	87	95	103	112	121	132										

Likelihood of Heat Disorders with Prolonged Exposure or Strenuous Activity

Caution · Extreme Caution · Danger · Extreme Danger


WHY
HUMIDITY
MAKES IT FEEL SO
HOT

L2 Water in the Atmosphere

August 182, 2025

HUMIDITY

the amount of water vapor in the air at any given time

DEW POINT

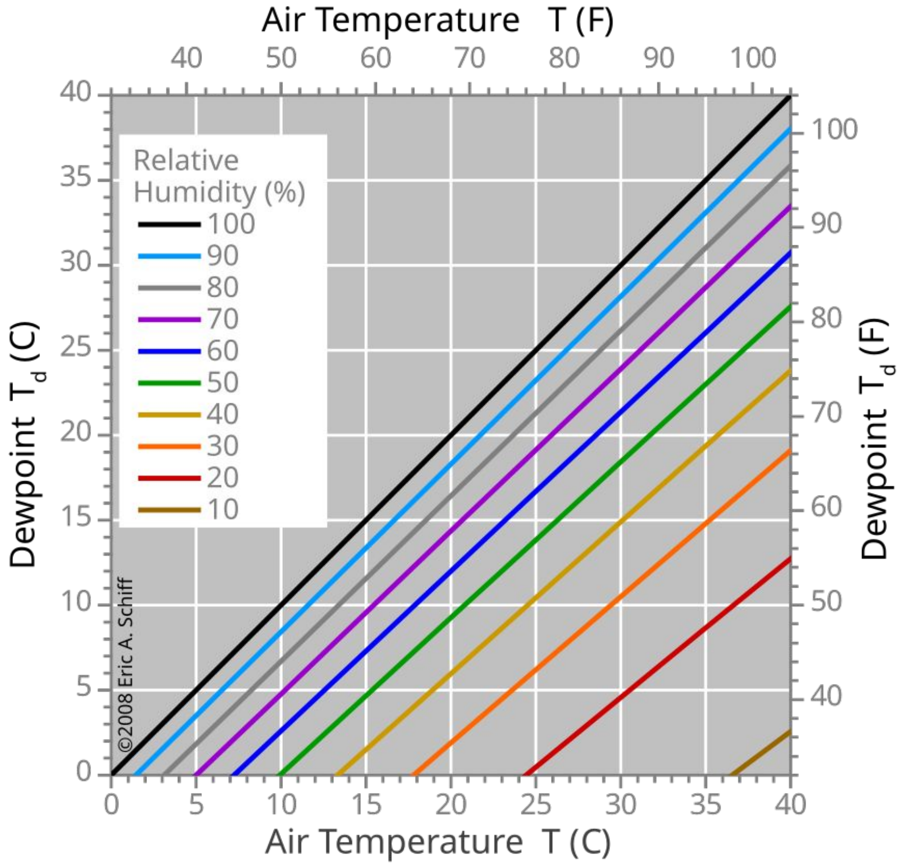
the temperature at which the amount of water vapor in the air would start to condense on any given surface. Uncomfortable: between 55-65

RELATIVE HUMIDITY

humidity expressed as a percent of how much water the air could carry at that given temperature

L2 Water in the Atmosphere

August 182, 2025



- at 100% humidity, the dew point is the temperature
- a dew point of 50 (F) is the same AMOUNT of water vapor no matter what the temp

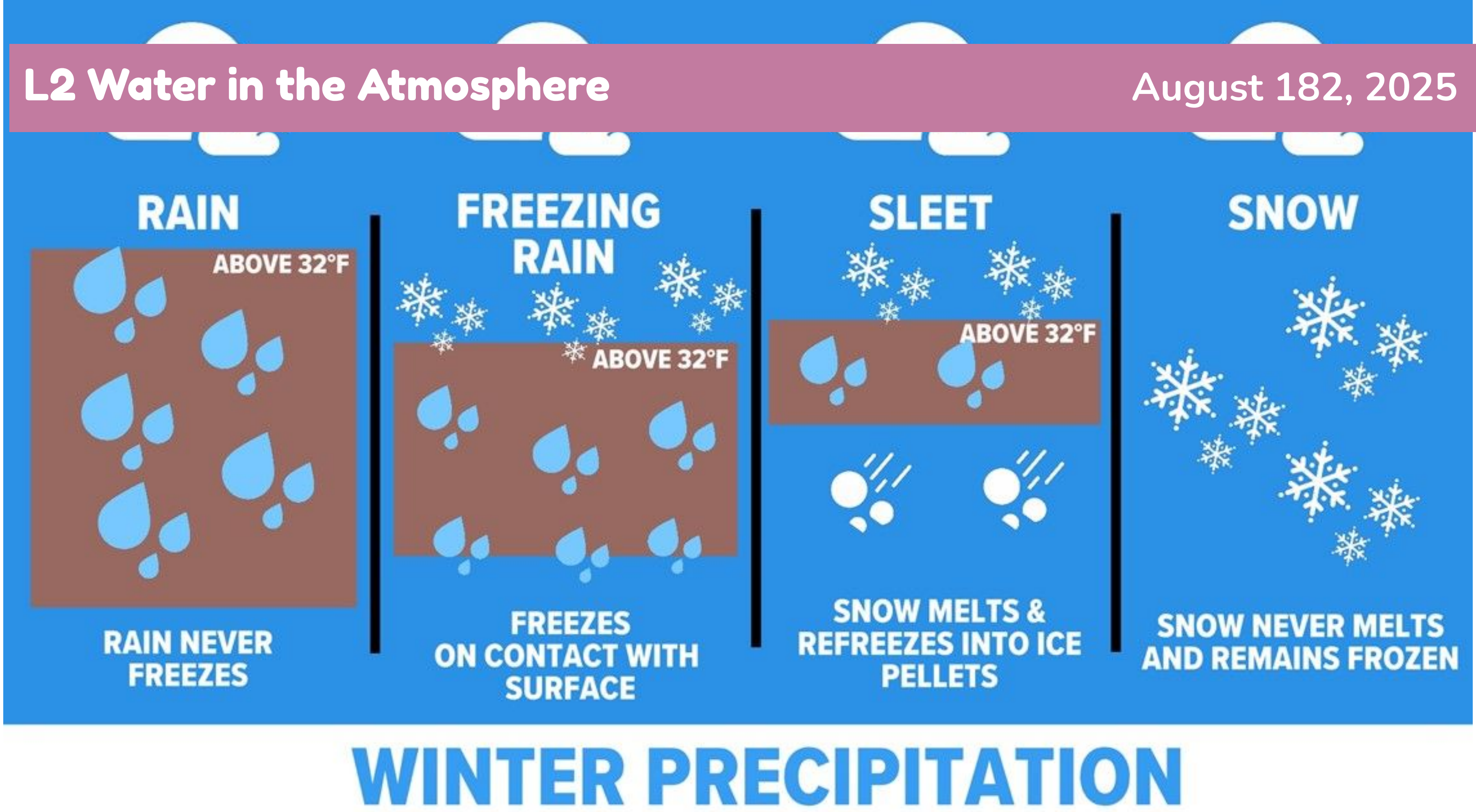
L2 Water in the Atmosphere

August 182, 2025

PRECIPITATION	PERCEPTION
droplet	you can see the clouds, might look hazy
mist	air looks hazy, but you can't feel it on your skin
drizzle	you can see the ground or your clothes getting damp, maybe you can feel it
rain	see the raindrops, feel them hit your skin, hear them hitting objects

L2 Water in the Atmosphere

August 182, 2025



Rain: never freezes

Freezing Rain: passes through a warm layer of air, melting, but returns to freezing temperature when it hits the ground

Sleet: passes through a warm layer of air, melting, but has enough time to refreeze into little icy bullets before hitting the ground

Snow: ice crystals form into snowflakes and stay frozen

February 2, 2026

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Cold Air	Warm Air
MORE DENSE	**LESS DENSE**
SINKS	**RISES**
IS COLD	**IS WARM**
HIGH PRESSURE	**LOW PRESSURE**


Broadleaf Stonecrop
Sedum spathulifolium
© Greg Dahlman

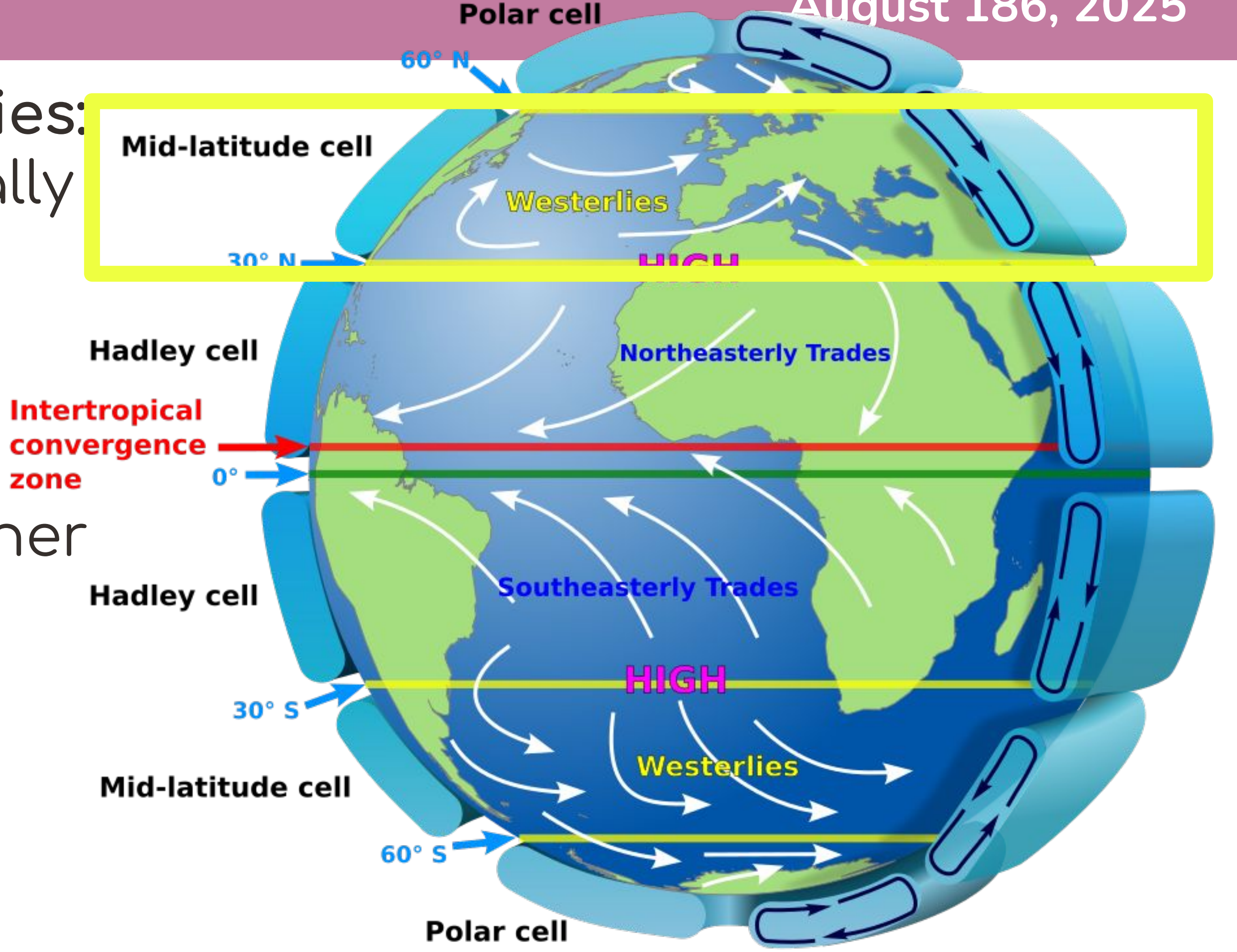
Thermal Properties

air is influenced by the region it's formed over

- heat transfer will make the air the temp of the region
- bodies of water have a stable temp
- land temp changes drastically between day and night

Prevailing Westerlies: air masses generally move from west to east

Jet stream: high-speed wind at higher altitudes



L3 Air Masses

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Rules for Fronts:

- Two or more air masses: Warm air and cold air collide
- Create clouds and or precipitation: WEATHER!!!
- Humidity in warmer air mass condenses
- Factors affecting fronts:
  - Temperature (how hot, how cold, the difference)
  - Humidity
  - Size of air mass
  - Speed and direction of movement
  - Region that it's going over


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WHAT ARE
CYCLONES?

L4 Lesson Check

August 196, 2025

Question 3:

- What data does a weather station need to collect?
- What instruments are used to collect that data?

San Jose, San Jose International Airport (KSJC)

Lat: 37.35917°N **Lon:** 121.92417°W **Elev:** 59.0ft.

Partly Cloudy

55°F

13°C

Humidity	66%
Wind Speed	N 0 MPH
Barometer	30.24 in (1024.04 mb)
Dewpoint	44°F (7°C)
Visibility	10.00 mi
Last update	12 Feb 10:00 AM PST

Pre-Lab Questions

1. Why does the tea in the cups lose thermal energy faster than the tea in the teapot? Give three reasons. Remember to use academic vocabulary!

$ WORD BANK $

Heat	Conduction
Temperature	Insulator
Thermal Energy	Conductor

Materials

2 styrofoam cups

2 thermometers

Hot water

Aluminum foil

COPY ALL THIS INTO YOUR NOTEBOOK :)

Procedure Summary

Experiment 1: Volume

Pour different amounts of water into cups with water of the same temperature and monitor temp for 10 minutes.

Experiment 2: Lid

Pour the same amount of water into two cups, but cover one with aluminum foil. Monitor temp for 10 minutes.